Sleep duration and next-day word recall: a synthetic preprint

Abstract

We studied how sleep duration relates to next-day word recall in a synthetic cohort of 240 adults. Longer sleep was associated with better recall, and the association remained after adjusting for age.

1 Introduction

Sleep is thought to support memory consolidation. Earlier laboratory studies reported mixed results, partly because sleep was measured by self-report. This fixture describes an invented study used only for tests.

2 Methods

Participants wore an actigraphy device for seven nights. Each morning they studied a list of 40 words and recalled as many as possible after thirty minutes. We fitted a linear mixed model with participant as a random effect.

3 Results

Participants who slept more than seven hours recalled 18% more words than those who slept less than five hours. The difference was smaller in participants older than sixty years. No association was found for reaction time.

4 Discussion

The association between sleep and recall does not establish causation, because the study was observational. Future randomized studies should manipulate sleep duration directly.